Automated Analysis of Calcium Transients in Engineered Cardiac Microtissues

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Abstract. Engineered Cardiac Microtissue (eMT) recordings provide spatiotemporal information on cardiac calcium dynamics, yet current analyses primarily rely on global intensity signals and overlook spatially resolved motion behavior. We introduce a fully automated, phase-aware framework for spatial motion quantification in eMT calcium imaging videos. The proposed pipeline integrates tissue segmentation, event-based phase separation, and optical flow–based tracking to compute region-specific metrics of velocity, directional consistency, and spatial heterogeneity during contraction and relaxation. We evaluate the framework on eMT recordings acquired under nifedipine and verapamil treatment across five concentrations and five maturation stages. Across all conditions, contraction velocities exceeded relaxation velocities (median ratio 2.7×, $p < 0.001$). Verapamil produced a significantly steeper dose-response reduction in contraction velocity compared to nifedipine ($p = 0.009$). Spatial analysis further revealed persistent regional velocity gradients and a dose-dependent reduction in within-tissue heterogeneity under verapamil treatment. Maturation analysis revealed an increase in directional dispersion with tissue age ($p = 0.002$), suggesting progressive spatial differentiation of contractile motion. These findings demonstrate that phase-separated, spatially resolved motion quantification enables automated pharmacodynamic phenotyping beyond conventional signal-based analysis.

Keywords: Engineered Cardiac Microtissue · Calcium Imaging · Optical Flow · Spatiotemporal Motion Quantification · Computational Phenotyping

1 Introduction

Engineered Cardiac Microtissues (eMTs) have emerged as a human-relevant model for studying cardiac physiology and drug response [25]. As animal models do not always accurately reflect human cardiac dynamics [8, 23], eMT-based imaging platforms have become increasingly important for preclinical evaluation [10, 2].

Calcium transient analysis in cardiac tissues has traditionally relied on fluorescence intensity traces, from which temporal features such as amplitude, rise

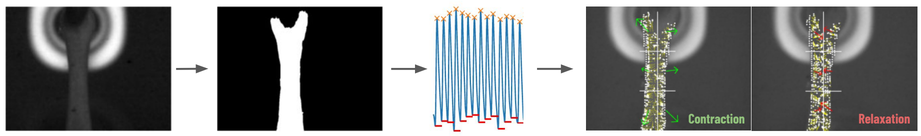

Fig. 1. Overview of the proposed motion analysis pipeline. From left to right: a fluorescence frame of the eMT, the segmentation mask, and the extracted calcium intensity trace. Individual beats are detected from the trace and divided into contraction (rise) and relaxation (decay) phases. Optical flow is computed within the segmented region for each phase, yielding phase-specific velocity and directional metrics across a 3×2 spatial grid.

time, decay time, and beat rate are extracted [22, 28]. These analyses are commonly performed either at the single-cell level or using region-averaged tissue signals, providing valuable electrophysiological characterization [26, 19]. However, such intensity-based metrics do not explicitly quantify spatially resolved motion dynamics or phase-specific mechanical differences across the tissue surface.

Optical flow–based motion tracking has been applied in prior work to characterize calcium wave propagation in engineered cardiac tissues [9, 5], combining segmentation, fluorescence signal processing, and sparse optical flow to quantify propagation velocity, traveled distance, and directional consistency during signal rise and decay phases. However, the prior framework emphasized propagation characteristics and directional consistency, whereas statistical dose-response comparisons and quantification of within-tissue spatial heterogeneity remained unexplored.

In this work, we apply an optical flow–based motion analysis framework to a new dataset of eMT calcium recordings acquired under two drugs (verapamil and nifedipine) across five concentrations and five maturation stages. We introduce three analysis contributions: (1) dose-response characterization with regression-based comparison of drug-specific velocity slopes; (2) quantification of contraction-relaxation velocity differences across all experimental conditions; (3) spatial heterogeneity metrics revealing dose-dependent changes in within-tissue motion uniformity; and (4) maturation-dependent directional analysis revealing age-related spatial differentiation of motion patterns. These analyses show that phase-aware mechanical quantification yields statistically interpretable pharmacological signatures beyond conventional calcium transient analysis.

2 Dataset

Engineered microtissues (eMTs) are three-dimensional cardiac constructs comprising iPSC-derived cardiomyocytes, fibroblasts, and extracellular matrix components (approximately 1 mm diameter, 6.8 mm length). Calcium dynamics were visualized using the genetically encoded indicator GCaMP6.

Fluorescence microscopy recordings of eMTs were acquired to analyze calcium-driven contractile motion. Each recording consists of a one-minute video containing 6000 grayscale frames at a spatial resolution of 720 × 540 pixels, acquired with a Basler acA720-520um at a sampling frequency of 100 Hz. The excitation wavelength is 470 nm. Intracellular calcium dynamics are visualized by changes in fluorescence intensity, with brighter regions indicating elevated calcium concentration during contraction. The bright halo visible around the tissue boundary in fluorescence frames originates from autofluorescence of the support frame and does not affect the analysis, as all motion metrics are computed exclusively within the segmented tissue region.

Two calcium channel blocking drugs were tested: nifedipine and verapamil. For each drug, recordings were obtained across five concentrations (0.02, 0.1, 0.5, 1.0, and 10 µM), along with a baseline condition without drug exposure. In addition, maturation-dependent recordings were collected at five developmental stages (days 11, 15, 19, 23, and 27), with day 27 representing fully mature tissue.

The dataset, therefore, includes two experimental axes: (1) drug concentration variation (n = 18 recordings per drug) and (2) tissue maturation stage (n = 15 recordings per drug). All recordings were analyzed using the same preprocessing and motion-extraction pipeline shown in Fig. 1.

3 Method

3.1 Data Preprocessing

To standardize intensity distributions across sequences, each frame was clipped at the 0.35th and 99.65th percentiles and linearly rescaled to the range [0, 255]. Percentile-based clipping reduces the influence of extreme outliers and improves robustness to illumination variability [24, 7]. For signal analysis, the image stack was normalized per frame and smoothed using a three-dimensional Gaussian filter with spatial $\sigma = 3$ pixels and temporal $\sigma = 4$ frames. Spatiotemporal smoothing suppresses high-frequency noise while preserving coherent contraction-related intensity transitions, which is critical for stable optical flow estimation [13, 16].

3.2 Segmentation

Binary tissue masks were generated using a pretrained U-Net architecture [21]. The model was trained on an independent set of eMT recordings acquired under identical imaging conditions but excluded from the present analysis. During inference, frame-wise masks were produced without manual interaction. Post-processing was applied to enforce topological consistency. Predicted masks were binarized, internal holes were filled, and small disconnected components were removed. Components smaller than 10% of the largest connected region were discarded to preserve the dominant tissue structure [1, 17]. To ensure temporal consistency across frames, we introduced a geometry-constrained mask quality scoring mechanism. For each frame, a spatial score was computed based on the concentration of foreground pixels within a central vertical band (35% of frame width), penalizing pixels outside this region and above the expected tissue origin. The highest-scoring mask in each sequence was selected as a reference template. Frames with normalized scores below 0.80 were replaced by the reference mask. This procedure acts as a temporal regularization step, reducing frame-to-frame segmentation drift and ensuring spatial alignment throughout the sequence [6]. Segmentation quality was assessed on three randomly selected sequences per condition via manual frame-level inspection (n = 78 frames), yielding a precision of 0.96 and a recall of 0.88.

3.3 Peak Detection

A global calcium fluorescence trace was extracted from each recording using a fixed reference tissue mask. For each frame, the mean intensity within the tissue region was computed, and the mean background intensity outside the mask was subtracted to suppress non-tissue contributions and illumination drift. The normalized calcium signal was computed as

$$\Delta F/F_0 = \frac{S_t - F_0}{F_0},$$

where S_t is the background-corrected tissue fluorescence and F_0 is defined as the median signal over the first 3000 baseline frames [20, 18, 11]. Median-based estimation was chosen for robustness against transient fluctuations and noise.

Contraction events were detected as peaks in the $\Delta F/F_0$ trace. To avoid noise-driven detections, peaks were filtered using a prominence threshold and a minimum distance constraint based on the typical inter-beat interval [27]. For each valid peak, contraction and relaxation phases were defined using the local minima before and after the peak. The rise phase was defined as the interval from the preceding minimum to the peak, and the decay phase as the interval from the peak to the following minimum. These phase intervals were subsequently used to compute the rise and decay durations and signal variability metrics.

3.4 Motion Quantification

Optical Flow Estimation Motion fields were estimated using the pyramidal Lucas–Kanade algorithm [14, 13, 4], implemented in OpenCV. For each detected contraction event, tracking points were initialized on a regular grid (spacing = 10 pixels) within the segmented tissue mask. Sparse optical flow was computed between consecutive frames through the full event duration (contraction onset to relaxation end) using a window size of 25×25 pixels, a maximum pyramid level of 3, and convergence criteria of 50 iterations with $\varepsilon = 0.001$. Tracking points that exited the tissue mask or image boundary were constrained to their last valid position to maintain trajectory continuity. A noise floor was estimated from optical flow displacements during inter-beat rest intervals (95th percentile of step magnitudes in quiet windows). Flow vectors below this threshold were excluded from subsequent analysis.

Phase-Separated Velocity For each contraction event, motion trajectories were analyzed separately during contraction (rise) and relaxation (decay) phases, as defined by the detected peak frame (Section 3.3). For each grid cell, a cumulative displacement vector was computed per trajectory as the vector sum of frame-to-frame displacements exceeding the noise floor. Displacement vectors were averaged across all trajectories within each cell, and regional velocity was defined as:

$$v = \frac{\left\| \frac{1}{N} \sum_{j=1}^{N} \mathbf{D}_j \right\|}{T_{\text{phase}}},$$

where $\mathbf{D}_j = \sum_t \mathbf{d}_{j,t}$ is the cumulative displacement of the j-th trajectory (excluding steps below the noise floor), N is the number of valid trajectories, and T_{phase} is the phase duration in seconds. Pixel displacements were converted using a spatial scale of 6.9 μm/pixel (Basler acA720-520um, pixel pitch 6.9 μm, no binning, 1× magnification).

Motion Direction and Consistency Motion direction was derived from the same cumulative displacement vectors used for velocity computation. Directional consistency across tracking points was quantified using the mean resultant length (MRL) from circular statistics [15]:

$$R = \frac{1}{n} \left\| \sum_{j=1}^{n} \hat{\mathbf{d}}_j \right\|, \tag{1}$$

where $\hat{\mathbf{d}}_j$ is the unit displacement vector of the j-th trajectory and n is the number of valid trajectories. The metric $R \in [0, 1]$ measures directional concentration, with $R = 1$ indicating perfectly aligned motion and $R = 0$ indicating isotropic directionality. Trajectories with net displacement below 0.5 pixels were excluded to avoid instability from near-zero vectors.

3.5 Spatial Analysis

The bounding box of the reference tissue mask was partitioned into a fixed 3×2 spatial grid (top/middle/bottom × left/right). Within each grid cell, contraction velocity, relaxation velocity, directional consistency (R), and mean motion direction were computed independently. Phase asymmetry was quantified as $v_{\text{contraction}}/v_{\text{relaxation}}$ per region. Within-tissue heterogeneity was defined as the range of regional velocities (maximum minus minimum across grid cells).

4 Results

4.1 Phase-Separated Motion Characteristics

Across all conditions, phase-specific velocity analysis revealed that contraction velocity was consistently higher than relaxation velocity. The median contraction-to-relaxation velocity ratio was 2.7× ($p < 0.001$), indicating systematically

Table 1. Summary statistics of contraction and relaxation velocities across drug conditions. Values are reported as mean ± standard deviation ($n = 3$ per condition). Ratio indicates contraction velocity divided by relaxation velocity.

Drug	Condition	n	Contr. vel. (mm/s)	Relax. vel. (mm/s)	Ratio
Verapamil	Baseline	3	0.109 ± 0.013	0.017 ± 0.010	7.7 ± 3.0
Verapamil	0.02 μM	3	0.171 ± 0.037	0.082 ± 0.013	2.1 ± 0.3
Verapamil	0.1 μM	3	0.162 ± 0.015	0.058 ± 0.006	2.8 ± 0.5
Verapamil	0.5 μM	3	0.124 ± 0.014	0.043 ± 0.005	2.9 ± 0.5
Verapamil	1.0 μM	3	0.058 ± 0.017	0.033 ± 0.015	1.9 ± 0.3
Verapamil	10.0 μM	3	0.042 ± 0.017	0.026 ± 0.013	1.7 ± 0.4
Nifedipine	Baseline	3	0.121 ± 0.013	0.030 ± 0.007	4.2 ± 1.0
Nifedipine	0.02 μM	3	0.128 ± 0.021	0.050 ± 0.020	2.7 ± 0.6
Nifedipine	0.1 μM	3	0.133 ± 0.014	0.043 ± 0.001	3.1 ± 0.4
Nifedipine	0.5 μM	3	0.136 ± 0.031	0.042 ± 0.005	3.2 ± 0.5
Nifedipine	1.0 μM	3	0.093 ± 0.044	0.030 ± 0.012	3.1 ± 0.4
Nifedipine	10.0 μM	3	0.080 ± 0.062	0.023 ± 0.008	3.2 ± 1.6

higher motion magnitude during the rise phase. Table 1 summarizes mean contraction and relaxation velocities for both drugs across concentrations. Baseline contraction velocities were comparable between verapamil (0.109 ± 0.013 mm/s) and nifedipine (0.121 ± 0.013 mm/s), while relaxation velocities were consistently lower, resulting in elevated velocity ratios at baseline. Directional analysis further validated phase separation: contraction and relaxation vectors were near-opposite in 90% of events (median angular difference = 162°), confirming consistent bidirectional motion across beats.

4.2 Dose-Dependent Response

Figure 2 shows regression analysis demonstrated a strong negative dose-response relationship for verapamil in contraction velocity ($r = -0.88$), corresponding to a 4.1× reduction between 0.02 μM and 10 μM. Relaxation velocity exhibited a similar but less pronounced decline (3.1× reduction). In contrast, nifedipine exhibited a weaker correlation with contraction velocity ($r = -0.47$), with a 1.6× reduction over the same concentration range. Relaxation velocity also showed a weaker decline (2.1× reduction). Direct comparison of regression slopes revealed significantly stronger suppression under verapamil for both contraction ($p = 0.009$) and relaxation ($p = 0.014$). These results demonstrate that phase-specific motion descriptors capture drug-dependent mechanical suppression beyond global intensity measures.

Figure 3 presents regional velocity patterns across six grid positions. Significant velocity differences were observed across top, middle, and bottom regions (Kruskal–Wallis, $p < 0.001$), with the pillar-constrained central region exhibiting the lowest motion magnitude. Verapamil induced a significant reduction in within-tissue velocity dispersion ($r = -0.84$, $p = 0.0001$), indicating progressive

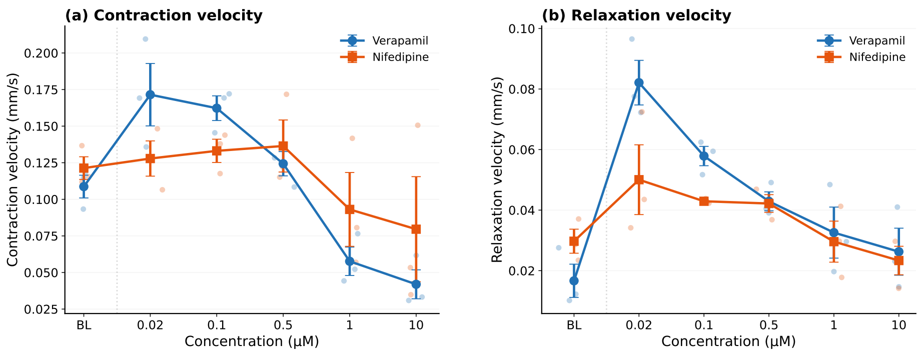

Fig. 2. Dose-dependent velocity suppression. Contraction velocity (a) and relaxation velocity (b) across five concentrations for verapamil and nifedipine. Points indicate individual recordings; lines show mean ± SEM (n=3 per condition). BL = baseline (no drug).

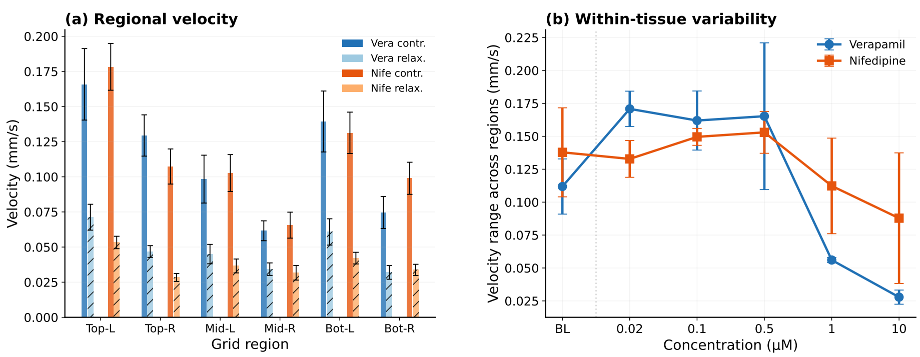

Fig. 3. Directional dispersion across maturation stages. Angular spread of contraction (a) and relaxation (b) motion vectors at five developmental stages (day 11–27). Higher values indicate greater directional heterogeneity across tissue regions.

spatial homogenization at higher concentrations. In contrast, nifedipine showed no significant trend in regional dispersion ($r = -0.23$, $p = 0.411$).

Mean principal flow direction remained stable across concentrations for both drugs, suggesting that pharmacological modulation primarily affects motion magnitude rather than global contraction axis. Directional consistency increased under verapamil during relaxation ($r = -0.58$, $p = 0.024$), further supporting dose-dependent spatial regularization (Fig. 3b).

4.3 Maturation Effects

Contraction velocity did not exhibit a significant age-dependent trend overall, though a moderate positive correlation was observed under nifedipine ($r =$

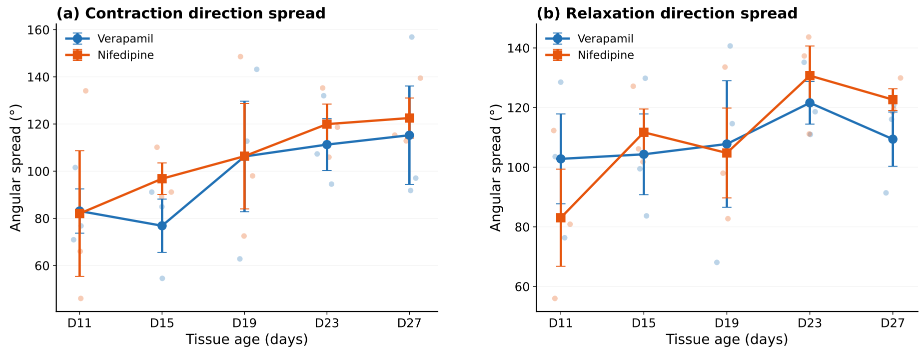

Fig. 4. Spatial organization of tissue motion. (a) Regional velocity across six grid positions by drug and phase. (b) Within-tissue velocity range as a function of concentration, quantifying spatial heterogeneity.

-0.58, $p = 0.024$). In contrast, directional dispersion increased with age during contraction for both nifedipine ($r = 0.55$, $p = 0.035$) and verapamil ($r = 0.52$, $p = 0.045$). During relaxation, a positive correlation with age was observed only under nifedipine ($r = 0.59$, $p = 0.0321$), indicating progressively more spatially differentiated motion patterns in mature tissues [12, 3](Fig. 4).

5 Discussion

Verapamil produced significantly greater velocity suppression and progressive spatial homogenization than nifedipine, demonstrating that the proposed framework captures drug-specific mechanical signatures. Maturation analysis revealed an increase in directional dispersion with tissue age, suggesting progressive regional specialization of contractile motion patterns during development.

The sample size per condition (n = 3) limits statistical power, particularly for baseline comparisons where high inter-sample variability was observed. Noise-floor estimation relied on inter-beat rest intervals, which may be insufficient in recordings with high spontaneous beat rates.

6 Conclusion and Future Work

We presented a fully automated, phase-aware framework for spatially resolved motion quantification in eMT calcium imaging. The pipeline captures drug-specific dose-response signatures, spatial heterogeneity patterns, and maturation-dependent directional changes that are inaccessible to conventional intensity-based analysis. Future work will validate this framework on larger cohorts and explore generalization to other fluorescence-based contractile assays for automated phenotypic screening in cardiac drug development.

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